

Resolving the Einstein, Podolsky, Rosen, Bohm, and Aharonov Paradox with Non-relativistic Quantum Mechanics... for Photons

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Abstract

The results of a previous analysis of the Einstein, Podolsky, Rosen, Bohm, and Aharonov (EPRBA) experiment for electrons presented a local-theory calculation of the correct quantum correlation results. This result was obtained through disentanglement, the correct use of the $SU(2)$ symmetry of the particles, a recognition that spinors are not 3-dimensional, an exact mapping of spinors onto three dimensions, and an inclusive path integral treatment of the unobservable perpendicular components of spin. Here, the same formalism is applied to photons which obey the same $SU(2)$ symmetry and map onto the same three dimensions. However, photons provide a deeper understanding of the wave function, disentanglement, and the 3-dimensional mapping. In the end, we arrive at the same conclusion: a correct handling of spin shows that standard quantum theory is local.

Keywords

EPR Experiment, Spinors, Nonlocality, Local Realism, Decoherence, Quantum Variables, Bell's Theorem, Photons

Bohm and Aharonov's version (Bohm 1957) of the Einstein, Podolsky, Rosen (1935) thought experiment is phrased as applying to electrons. Experimental tests have been conducted with electrons (Hensen 2015) and protons (Lamehi-Rachti 1976) but the predicted observations were, and continue to be, most practicable using photons. For that reason, most discussions of the effect are tested and described in terms of photons, and carried out with laser beams, polarizers, and photodiode detectors.

As discussed in the previous article (Stoller 2026), many of these experiments aim to explore Bell's Theorem (Bell 1964) and, through this, substantiate the general conclusion that quantum mechanics is nonlocal. Such conclusions are misunderstood because the Bell inequalities only apply to classically local theories. Bell's and related inequalities do not impose any condition on local variable quantum

theories. This is implicit in Bell's work, is stated explicitly by Clauser (1969), and demonstrated in our previous article (Stoller 2026).

A local variable quantum theory refers to a theory of non-commuting operators with spatially localized effects. A local quantum theory is one that only predicts local effects. This would be a theory whose expectation values for observations at any point only involve values at that point, or at points causally connected to the point of observation through the retarded effects of interactions, fields, or potentials.

Bell and CHSH-type (Clauser 1969) theorems tell us that the predictions of quantum theory cannot be reproduced by a local classical theory, which is a theory based solely on local interactions and classically commuting parameters. These theorems tell us nothing about the structure of quantum theories outside the classical realm.

Bohm (1952) presented his De Broglie-Bohm pilot-wave theory (Holland 1995, p.26) as a nonlocal classical theory that reproduces the results of quantum mechanics. Bell-CHSH theorems indicate that no such local classical theory can do the same. In Particular, Bell-CHSH theorems do not tell us whether or not there can exist a local non-classical theory that will reproduce the predictions of quantum mechanics. Fisher (2023) has recently pointed out that stochastic local theories can reproduce the predictions of quantum mechanics. Standard quantum mechanics is such a theory.

Bell-CHSH theorems do not tell us quantum mechanics is nonlocal. That conclusion has been inferred from various quantum descriptions that use nonlocal constructions. The claim that quantum mechanics is always nonlocal requires the additional claim that such nonlocal constructions are always required.

In our previous paper (Stoller 2026) we showed that nonlocal constructions are not required for Bohm and Aharonov's paradigmatic, massive, spin- $\frac{1}{2}$ particle, thought experiment. We do this by providing a causal, local construction that generates the observed results. We also showed that this does not conflict with Bell-CHSH inequalities. Our description uses the standard SU(2) spin representation in a path integral formulation that, we claim for the first time, treats spinors correctly.

In the current paper we extend this to experiments with photons. We show that nonlocal constructions are again unnecessary and that a locally constructed quantum theory is sufficient. That is, pure quantum mechanics, using local quantum variables, generates the observed quantum correlations, and does not have recourse to the acausal, nonlocal assumptions. Part of this demonstration rests on a careful understanding of the different meanings of similar ways the wave function can be written.

1 - Choice of the Wave Function

Quantum mechanics distinguishes between pure and mixed states. Pure states can be written as a single product of eigenstates; mixed states can only be written as a sum of products of eigenstates. Pure states exist in a single Hilbert space and can be represented by a single vector. Mixed states exist in a vector sum (irreducible sum) of Hilbert spaces where each space represents a different and exclusive reality.

A system in a pure state is in a single, certain configuration while a system in a mixed state is in a classically uncertain configuration. Classically uncertainty means that the observed configuration will

be one of many that are possible, or that many states are contemporaneously superposed and we are uncertain which actually represents the system. This ambiguity between what exists and what will be observed occurs in many quantum descriptions of a system's configuration. It is this ambiguity that gives rise to Everett's "Many Worlds" interpretation of quantum mechanics (DeWitt 1973, Varshovi 2025).

Quantum mechanical representations do not distinguish between the superposition of pure states and a mixture of pure states. That is, a state $\Phi = \phi_1 + \phi_2$ can represent a state where ϕ_1 and ϕ_2 are in the same Hilbert space, such as two electrons sharing a 2p orbital, or in different Hilbert spaces, such as two nonoverlapping electrons that are distinguishably separated. If the components are in the same space, Φ is a pure state. If they are in different spaces, Φ is a mixed state. The states may formally look the same, depending on the formalism used, but generate different predictions.

An entangled state is a special form of mixed state where the components exist simultaneously in the same Hilbert space, or a tensor product of Hilbert spaces, but cannot be expressed as a pure state. Entangled states mix separate states that exist at the same time within the same vector space. The components of entangled states can interfere because they are co-located, while the components of mixed states cannot because they are not and cannot overlap.

A wave function can also represent an ensemble, which is a sum of inequivalent wave functions defined at different times contributing probabilities to the predicted outcome of a repeated experiment. We distinguish ensembles from mixed states by looking for interference or transitions between the pure states. Ensembles lack transitions between their components because the components of ensemble states do not co-exist.

The behavior of quantum states depends on what they're designed to represent. Ensemble, entangled, and mixed states can be written as superpositions that are formally indistinguishable. They are distinguishable only when the state spaces of their components are identified.

If we cannot test for interference or transitions, and if we are uncertain whether we're looking at one system and or an ensemble of systems, then we cannot distinguish an ensemble from a mixed state. We may get the same results from calculations done in either representation, but we'll interpret the results differently.

Photons present a particular difficulty in this regard since they are not localized. This makes it difficult to determine whether a state function that represents a collection is a mixture of separate photons, a self-interfering entanglement represented by a single photon, or an ensemble of noninteracting photons.

Consider the typical pure product state $|+1, -2\rangle$ representing spin component of electron number 1 in a spin-up state and electron 2 in a spin-down state. The spatial components of these two electrons may be identical or overlapping and are left unspecified. When we define a mixed state's spin component $\Psi = 1/\sqrt{2}(|+1, -2\rangle + |-1, +2\rangle)$ as the spin component of two electrons sharing the 2p orbital in a triplet

configuration we're saying that quantum statistics requires this symmetrized form of the wave function in order to describe the state of both electrons co-located in space and time.

We cannot distinguish these electrons and we cannot observe their separate effects while they're in spatially overlapping states. This is a mixed state, not an ensemble, and the uncertainty is quantum mechanical not classical. That is, the electrons are in a combination of both states at once, but will only be seen in one or the other.

In contrast, when we write $\Psi = 1/\sqrt{2}(|+a,-b\rangle + |-a,+b\rangle)$ to represent the wave function of two electrons that have left the 2p orbital and separated to distinct locations a and b, then we can either be specifying a 2-electron mixed state or an ensemble of two 2-electron states. If we're specifying a mixed state, then the measurement at either location has a 50% probability of detecting an electron at that location in a + or – state, a zero probability of both electrons being in the same state at the same location, and a zero probability of neither electron being in a + or – state at either location.

But if we are specifying an ensemble, then different tests at different times will reveal different elements in the ensemble. In this case, observations might show there to be – states and + states at both locations at different times.

2 - Ensembles

To avoid the above-mentioned confusion in the case of photons, experimenters strive to limit themselves to observations of coincident pairs. That is, observations that involve only a single pair of correlated photons. In this way it is asserted that these experiments involve mixed state wave functions. These are called entangled wave functions because the state is a simultaneous mixture of two configurations and they are not a classically uncertain mixture of one configuration or the other.

In the following sections we argue that coincident pairs do not sufficiently distinguish multi-photon ensembles from single photon mixed states, and that experiments of the sort considered below likely involve a form of multi-photon ensembles.

An entangled wave function can be built by combining two distinct two-state systems. If $\phi(+1,-2)$ represents the pure-state of positive polarization at location 1 and negative polarization at location 2, and $\phi(-1,+2)$ represents negative polarization at location 1 and positive at location 2, then we can create the entangled state—sometimes called “the Bell state” which is the symmetrized polarization state—by creating the combination:

$$\Psi(+,-) = (1/\sqrt{2}) (\phi(+1,-2) + \phi(-1,+2))$$

When locations 1 and 2 overlap, and the spins at those locations cannot be distinguished, there is no conflict in saying the wave function Ψ represents a single, symmetrized system. We cannot make a measurement to determine the polarization states at locations 1 and 2 because we cannot distinguish the two locations, consequently the question of how information flows between the two locations has no meaning.

However, when these locations are separate and distinct we can determine the polarization states at the respective locations. Observing a polarization value of + or – at location 1 then immediately requires the – or + polarization to manifest at location 2, while the other polarization values, namely the values that match what was found at location 1, disappear or “collapse.”

Such an observation generates the paradox of how this instruction was conveyed between locations 1 and 2 when this effect is presumed to be instantaneous and the distance between 1 and 2 is finite but unbounded. This is the EPRBA paradox.

On the other hand, if $\phi(+1,-2)$ and $\phi(-1,+2)$ represent a mixture of two 2-particle systems separated in space and/or time, then what happens to or at one pair bears no relation to, and has no effect on, the other. There is no EPR paradox when the combined 2-particle wave function is interpreted as a mixture.

The problem, or paradox, of spontaneous collapse is also avoided by replacing $\Psi(+,-)$ with a pure state. Additionally, reinterpreting $\Psi(+,-)$ as an ensemble also avoids the instantaneous collapse paradox. In this case, we’re interpreting the wave function as a description of a system that is in one or another of the pure states, we just don’t know which. Observation just opens one of the doors of possibility; nothing “disappears” from behind the other door because there was nothing behind the other door in the first place.

Given that pure states can represent the required correlations, as we have shown, why do we bother with the entangled states? There are two parts to the answer to this question. The first is the claim that the entangled states are realistic because they give correct predictions. The second is that the pure states either don’t have the right properties, cannot reproduce quantum results, or fail to predict observations. We consider these reasons in what follows.

3 - Ensembles Don’t Collapse

Any mixed state can be expressed as a linear combination of the orthogonal eigenfunctions of some observable. When a measurement of this observable is made on a system that is in this mixed state—which presupposes measurement using an instrument that measures these eigenstates—only one of the eigenvalues is recorded. Since all the components of a mixed state are presumed to be simultaneously present—as the mixed state is a superposition of states—all the other eigenstate contributions “instantly” vanish through the acausal mechanism we call “collapse.” This collapse is instantaneous over all distances, as far as the formalism is concerned.

This “collapse of the wave function” presents a problem to those who would ascribe reality to all those components that disappeared. It must be noted that this feature of quantum mechanical state evolution is what makes quantum mechanics so widely applicable: as long as you provide the right transition probabilities, the quantum formalism doesn’t care how the state evolves. The formalism simply waves away the question of how the state evolves as if it were magic!

Here again, the observations of Mineev et al. (2019) on superconducting, artificial, three-level atoms demonstrate that there are unknown mechanisms behind the collapse. This indicates that, at least in these cases, the quantum state has not been completely described. This should not surprise us since the linear quantum formalism, relying on the linear Schrodinger equation, does not apply to nonlinear dynamics.

Besides the existence of unrecognized mechanisms, such as we describe here in section 12 between the polarization and the polarizer, we argue that there is no collapse at all. This is the case for the ensemble and the mixture, where the wave function's components do not interfere and there is no tunneling between them. In these cases, the emergence of one component from many simply reflects that the system being observed was only composed of this one component from the start.

A combination of polarized photons has ensemble properties in that the photon polarizations interfere but do not interact: they add when superimposed but they are not changed. In theory, they can be separated and then recombined and restored to their initial values. This means that when an observing device, such as a photomultiplier tube, interacts with one photon in a group of photons, it removes that photon's energy and polarization from the group without changing the contributions from the other photons.

If the subtraction of a photon or pair of photons is equal to the addition of such photons with negative phase components, then the registration of photon in a mixed state is a process of selection of one among several, rather than a registration of one and a collapse of all others.

The "collapse," in this case, is just a collapse of our ignorance regarding which of several states we're detecting. Ignorance collapses instantly across any distance; it is not limited by the speed of light. This behavior conflicts with nothing.

4 - Confusing Nonlocality and Acausality

Authors often fail to clarify the difference between nonlocality and acausality. Nonlocality means the effects at one location are determined by causes at separate locations without any conditions on time or mechanism. Newton's theory of gravity is a nonlocal theory as it expressed the force between two bodies in terms of their masses and a separation defined by two different locations.

Acausality means that either there is no cause for an effect or observation, there is no mediating mechanism connecting the cause to the effect, or the precipitating event acts instantly and with or without a causal connection. Linear quantum mechanics is an acausal theory in the sense that it does not attempt to describe systems during periods of nonlinear time evolution and, instead, just works with states constructed *ab initio*.

Bell (ref) defines causality as a theoretical structure that correlates actions with events in their past light cone. This only enforces a context for time ordering but does not provide or require any mechanism. Such a definition of causality allows for action-at-a-distance as long as the action occurs in the rearward light cone of the induced effects.

Linear quantum mechanics is more a structure for theories of correlation than a causal theory of anything. It gives us transition amplitudes but not transition mechanisms. Causal explanations require more refined theories, such as quantum electrodynamics.

Whenever the actions at a point are due to many time-varying sources, whose simultaneous effects are determined by their instantaneous locations, and are then summed at the same time as the observed effect, the action is being described both nonlocally and acausally. When the retarded nature of intermediary effects is taken into account, and there is some mechanism of communication such as a field, the description becomes causal, but it remains nonlocal.

The interference effects of multiple paths that converge at a point of observation, as demonstrated in how quantum mechanics treats the double-slit experiment, are nonlocal but causal. The collapse of the wave function, presumed to occur in many quantum mechanical superpositions, is nonlocal and acausal.

Effects whose predicted values involve space-like separated wave function components are nonlocal. When these occur within a region where wave functions spatially overlap, such as the transition between atomic orbitals or through a transistor, the effects are local within that location. However, if they are still lacking a mechanism, they remain acausal. That is not to say the phenomena cannot have a cause, rather that quantum mechanics need not, and usually does not, provide one.

5 - Quantum Local Variables

There is a general confusion about the meaning of local variables (Wiseman 2006, Vaidman 2019, Fisher 2021, Brown 2016) that results from failing to distinguish between theories involving classical or quantum parameters. Classical variables are c-numbers that commute, while quantum variables are q-numbers or operators that may not commute. This confusion is reflected in this passage by Abner Shimony (1986, p.185):

“A hidden variable theory, which maintains that a complete specification of a physical system requires the supplementation of the quantum state, the supplementary information being commonly known as ‘hidden variables.’ If this avenue is taken, then quantum indefiniteness can be understood epistemically, as incomplete knowledge of the details concerning the system. Quantum mechanical probabilities can then be interpreted in the same way as probabilities of classical statistical mechanics when some physical constraints are imposed.”

The above statement is definitively false with regard to the use of quantum hidden variables implemented here. We are not supplementing or replacing quantum variables with c-number variables, rather we are correctly managing spinor q-number variables. Neither are we supplementing the quantum state, instead we are handling existing spin variables correctly. As we understand these spin variables, they are “hidden” not because they give rise to some new variable, but because they don’t exist as c-numbers in three dimensions. Their pre-measurement values are unknowable and, even after measurement, these c-number values are not “real” in the sense of being intrinsic to the system being observed.

Q-number spin variables are not, as Bell (1966) describes, “‘hidden’ because if states with prescribed values of these variables could actually be prepared, quantum mechanics would be observationally inadequate.” Bell does not clearly state that the hidden variables he’s describing are classical, but this is what we must conclude because all theories of classical hidden variables satisfy the Bell-CHSH inequalities, whereas quantum theories violate these inequalities.

Quantum spin variables can behave in ways that disobey classical statistical mechanics. In what is probably the simplest example, classical statistics requires the average (or expected) value of a sum of observations to be equal to the sum of the average values of the observations. Quantum mechanics will violate this requirement whenever the observations involve non-commuting operators whose eigenvalues relate to different eigenvectors in the Hilbert space.

The expectation values of such incompatible observables require different vectors in the same Hilbert space, and so are not averaged over the same ensemble of classical states. This is a fundamental requirement for classical statistics, and quantum mechanics violates this, which is shown in following example. This deviant quantum behavior underlies the Bell-CHSH inequalities.

If A_i and B_i are the predicted results over a set $\{i\}$ of observations interpreted according to a classical theory, where $i=1,2,...N$, and $\{A+B\}$ represents the average of their sum, then

$$\{A + B\} = \frac{1}{N} \sum_{n=1}^N (A_i + B_i) = \frac{1}{N} \sum_{n=1}^N A_i + \frac{1}{N} \sum_{n=1}^N B_i = \{A\} + \{B\}, \text{ for any } A \text{ and } B.$$

In quantum mechanics the values of $\langle A \rangle$ and $\langle B \rangle$ depend on the state of the system. For these values to be known with certainty A and B must be compatible, commuting observables: $[A,B] = 0$. If they are incompatible observables, then they will both only have certain values when evaluated with respect to different eigenstates. If $|\Psi\rangle$ is an eigenstate of A , $|\phi\rangle$ an eigenstate of B , $[A,B] \neq 0$, and $|\Psi\rangle \neq |\phi\rangle$, then

$$\langle \Psi | A + B | \Psi \rangle \neq \langle \phi | A + B | \phi \rangle \neq \langle \Psi | A | \Psi \rangle + \langle \phi | B | \phi \rangle .$$

Or in brief, when A and B are noncommuting operators, $\{A+B\} \neq \langle A+B \rangle$, and nothing can be done to restore this to an equality.

Because of this discrepancy between classical and quantum expectation values, where the classical expectation value satisfies Bell’s inequality and the quantum expectation value violates it, Shimony (1986, p.190 and p.200) concludes:

“Hence no hidden variables theory satisfying relativistic locality (as construed above) can agree with all the predictions of quantum mechanics... The evidence is overwhelming against the family of local hidden variables theories...These results can be stated with great confidence.”

The fully quantum, relativistically local calculation of section 12 shows this conclusion is incorrect as it fails to make an exception for quantum mechanics itself as a relativistic, local hidden variables theory.

Einstein's original German phrase for what has been translated as "spooky action at a distance" is "spukhafte Fernwirkung," which he used in a 1947 letter to Max Born. We suggest “ghostly action at a

distance” would be a better translation. Einstein was not “spooked,” and our translation suggests that he was pointing to an unrecognized ghost in the quantum machine.

This ghost is the quantum spinor which, when treated correctly as we have done, provides the missing mechanism connecting the particle’s state with the state of the detector, and from that the correlation between detectors. This mechanism makes it unnecessary to invoke nonlocal, acausal effects. It stands to reason that the information that correlates the spins of remotely separated entities is embedded in the local values of the spins themselves.

6 - Single Photon Entangled States May Be Unrealistic

Arthur Zajonc (2003, p.S-3) notes,

“Unlike other observables in quantum theory, there is no Hermitian operator that straightforwardly corresponds to position for photons. Thus, while we can formulate a well-defined quantum-mechanical concept of position for electrons, protons and the like, we lack a parallel concept for the photon and similar particles with integer spin. The simple concept of spatio-temporal location must therefore be treated quite carefully for photons.”

In the previous article we argued that there was no reason to model the free particle situation of two separate and correlated fermions with the anti-symmetric, entangled spin-state given by the 2-particle fermionic bound state. Without a compelling reason for using the entangled state that leads to a prediction of action-at-a-distance, which is what this wave function collapse amounts to, this entangled state should be considered unphysical.

Photon states are more difficult to describe than massive particle states because photons can interact with anything, do not form localized bound states, and can only be contained through reflections that cause decoherence and change phase and polarization relationships. Note also that a photon condensate is not a localized bound state (Stein 2019).

One cannot redirect photons without employing reflection or absorption and re-emission, which leads to decoherence and ensuing phase and polarization randomization. Superpositions of photons with random phase relationships are equivalent to a mixture (Greenstein 2016, p.197), or what I’m calling “a simultaneous mixture.” Mixed-state wave functions no longer behave like superpositions.

We cannot assign a definitive location to a photon. We can create a situation where we know there to be one photon within a volume with some volumetric uncertainty, but reducing this uncertainty to zero is not possible. This raises questions about the form of the wave function that results when we combine photons.

When we “add together” two photons of uncertain location, how do we know if we’ve created a superposition or a mixture? Most optical superpositions describe beams not individual particles. When we contain photons within a reflecting cavity, even just a single photon, we lose phase and polarization information.

The general test of a superposition is the generation of interference, but a single photon, given by a wave function with a single term, can be broken into two terms by a basis rotation and can then interfere with itself. We believe a better test of entangled superposition is observable evidence of the necessary measurement-induced collapse. This need not be nonlocal or acausal, though it is both in the case of space-like separated components. In Appendix-I we suggest two experiments that would demonstrate the reality of nonlocal, acausal wave function collapse in situations such as we're concerned with here.

Glancy and Vasconcelos (2008) discuss several methods to produce superpositions of optical coherent states. Each method is described using different formalisms resulting in different descriptions, such as semi-classical, first, and second quantized. These techniques are difficult to realize and remain largely theoretical. And while these states do exhibit quantum properties, the question of whether they're composed of discrete photons or represent a single quantum state is unclear. Questions of the Bose-Einstein condensate nature of light continue to be explored (Kraus 2025).

The perfect and enduring anti-correlation we see between pairs of photons and pairs of electrons, which persists despite the fragile and easy decoherence of the entangled states, argues against entanglement and collapse as the mechanisms of anti-correlation between pairs.

The continued belief that the entanglement of two, single-photon states is the explanation for the correlations, as manifest by the probability of transmission of light through relatively oriented polarization filters, is based on the Bohm and Aharonov (1957) argument that pure states (i.e. nonentangled states) don't conserve angular momentum, and the Bell-CHSH theorems that are believed to prohibit a local description.

However, pure states do conserve angular momentum and Bell-CHSH related theorems do not prohibit a local description, as is revealed from a careful analysis of these theorems, and through the counter-example provided here.

The conservation of angular momentum for pure states follows from the invariance of the expectation value of spin correlation of correlated particles under an arbitrary rotation. This invariance is demonstrated by showing the following correlation has the value of -1 independent of a general 3-dimensional rotation $\boldsymbol{\theta}$:

$$\langle S_z(1)S_z(2) \rangle = \langle +_1 | \langle -_2 | U^\dagger(\boldsymbol{\theta}) S_z(1) S_z(2) U(\boldsymbol{\theta}) | +_1 \rangle | -_2 \rangle = -1$$

Here $|+_1\rangle$ and $|-_2\rangle$ are the spin-up and down kets of particles 1 and 2, relative to some z-axis. The z-component of spin for a particular particle is given by the expectation value $\langle S_z(1) \rangle$ and $\langle S_z(2) \rangle$ for particles 1 and 2. The generator of arbitrary rotations is given by $U(\boldsymbol{\theta}) = \exp(-i/2(\boldsymbol{\sigma} \cdot \boldsymbol{\theta}))$, where $\boldsymbol{\sigma}$ are the Pauli matrixes $\sigma_x, \sigma_y, \sigma_z$, and $\boldsymbol{\theta}$ are rotations $\theta_x, \theta_y, \theta_z$ around the x, y, and z axes. This calculation is presented in detail in the previous work.

7 - Photon Ensembles Appearing as Superpositions

The fundamental source of nonlocal acausality in the EPRBA paradox is the description of the system using an entangled wave function. This wave function specifies probabilities for certain correlations across distances separated by space-like intervals or, more accurately, instantaneously manifesting nonlocal correlations.

The argument for entanglement is either that we've intentionally created it, or that it arises due to the symmetric or anti-symmetric nature of identical particles. However, wave function symmetrization has no physical consequence until a measurement is made and entanglement only persists as long as the system can be isolated and decoherence avoided. In Appendix II of our previous work, we demonstrate that experiments based on the creation of entangled photon states fail to discriminate between their use of entangled versus mixed states.

All situations where symmetrization has measurable consequences—paradoxical, nonlocal, acausal, ambiguous situations of the EPRBA sort excepted—involve spatially overlapped identical particles (or composites). The Lamb shift is an obvious example. Wherever entanglement appears describing spatially separated components, the measurement-induced observation of only one component requires the theoretical assertion of the collapse of the wave function.

“In order to understand why interference effects occur in some cases and not in others, we need to recall that interference is the physical manifestation of the intrinsic indistinguishability of the sources or of the photon paths. If the different possible photon paths from source to detector are indistinguishable, then we have to add the corresponding probability amplitudes before squaring to obtain the probability... On the other hand, if there is some way, *even in principle*, of distinguishing between the possible photon paths, then the corresponding probabilities have to be added and there is no interference.” — **L. Mandel** (1999)

Based on the above principle of quantum mechanics, we suggest the following clarifying principle: wave function symmetrization and anti-symmetrization is only required when “identical” particles cannot be identified, which only occurs when their physical wave functions overlap. We suggest that wave functions lose their symmetrization when their identical constituents become separated by space-like intervals.

Our argument, explained in section 11 for this case, is that the entangled wave function is a combination of a physical product state and a simultaneous mixture. This requires no change in the formalism. It follows from this principle that a simultaneous mixture exhibits no wave function collapse and no acausality.

Arguments based on environmentally induced decoherence reach the same conclusion but require measurable environmental interactions. We claim vacuum fluctuations can cause decoherence of nonlocal entanglement. Such terms are not “real” in a directly observable sense; they arise from higher order terms in the renormalization of a wavefunction described using a propagator formalism. Vacuum

fluctuation induced decoherence is an actively discussed topic (Gundhi 2023, 2025), but distinctions between nonlocal entanglement and local superposition are still unclear.

From a theoretical standpoint it is simpler to recognize the effects of symmetrization to be dependent on the spatial overlap of the entangled terms that appear in the wave function. This is the same overlap that underlies quantum tunneling and electron band theory which both depend on some overlap of the spatial wave functions.

Applying this suggestion, we reinterpret the physical superposition, as generated by symmetrization for disjoint systems, as a mixed state that represents our ignorance of how the symmetrized terms decoupled. We reinterpret this mixed state contribution—assuming measurement or decoherence has not washed it away in the meantime—as a superposition at such a time when the disjoint systems are recombined and indistinguishability is regained.

If we don't do this, and instead insist that the superposition persists for space-like, disjoint separations of the system's components, then we find ourselves in the position of claiming that all particles are perpetually entangled in a universal wave function. This creates the belief that everything is always connected, a notion embraced by mystically inclined nondualists (Chopra 2019 [popular commentary], Laudisa 2025). While existentially meaningful, this claim is measurably meaningless.

The belief that certain remote correlations are the result of nonlocal, acausal relationships applied to entangled quantum systems, and that this precludes all forms of local variables, is supported by a misunderstanding that Bell-CHSH inequalities. This has led Bancal (2012) to suggest the observation of remote correlations could be due to some form of superluminal connection and that,

“Quantum non-locality must necessarily relate discontinuous parts of the universe that are arbitrarily distant. This gives further weight to the idea that quantum correlations somehow arise from outside spacetime, in the sense that no story in space and time can describe how they occur.”

However, as we show here, there is no such need for descriptions of this sort. In the words of E. T. Jaynes (1980, p.42), we need not resort to “medieval necromancy”.

8 - Pure-State Wave Functions

A wave function can represent a single object or an ensemble of things, and the formalism of quantum mechanics does not distinguish between them. The difference is determined by what the wave function is intended to represent or by how we construct it.

Given a continuous or discrete Hilbert space we can assign elements of this space to a wave function in order to represent an integral “thing.” We call this a pure state and assign quantum numbers to it.

In the case of two correlated spins, as claimed in our previous paper's analysis of the EPRBA situation, each component is in a pure spin state. These pure spin states are defined with respect some angle in 3-dimensional space—usually a measurement angle—with respect to which their spin is parallel or anti-parallel (labeled + or –) at a specified location.

$\phi(\pm j) \equiv \text{spin } \pm \text{ with respect to some fixed angle at location } j.$

In the case of spin, $\phi(\pm j)$ are basis states in a 2-dimensional Hilbert space and there is an independent Hilbert space for each particle.

The product of two wave functions, each representing a particle, is itself a pure state in the combined Hilbert space of the two spins. The pure, two-state component wave functions are separable; they are formed as the product of two, orthogonal Hilbert spaces. All of these two-state wave functions represent correlated spins.

$$\phi(+1,-2) = \phi(+1) \otimes \phi(-2), \text{ and } \phi(-1,+2) = \phi(-1) \otimes \phi(+2)$$

Where $\phi(+1,-2)$ means + spin of particle 1 at location 1, and – spin of particle 2 at location 2. The states $\phi(\pm j, \pm k)$, where $j \neq k$, form the basis states in a 4-dimensional Hilbert space. There is an independent Hilbert space for each particle location j and k .

Pure states for a given particle can be expressed using different bases, and the basis components in one base are generally not orthogonal to the basis components in another. That is, any state in one Hilbert space can be represented as a combination of base states in an equivalent Hilbert space. Each basis element, or vector, in any one Hilbert space can be written as being composed of a linear combination of basis vectors in another basis of the same Hilbert space.

How one maps one basis to another is determined by the algebra of the space. This algebra determines how one “rotates” from one basis to another. Such a rotation algebra is closed and complete; different bases are isomorphic. For electron spins and photon polarizations, rotations between different spin or polarization bases are governed by the SU(2) algebra.

9 - Entangled Wave Functions

Because the Schrodinger equation is linear, all linear combinations of solutions are also solutions. Linear combinations of pure states are called entangled states. When such combinations are not a rotation of the basis vectors, they are no longer pure states. $\psi(\pm)$, given as a linear combination of product states, is such an entangled state:

$$\psi(\pm) = \phi(+1,-2) \pm \phi(-1,+2)$$

Each $\phi(\pm i, \pm j)$ components is a pure state that's represented by a single ray in a multiplicatively combined Hilbert space. Such rays present a complete specification of system properties.

We can interpret this wave function ψ in one of three exclusive ways. Because the formalism looks the same in each case, we must provide additional specifications or perform additional experiments in order to distinguish them.

Entangled State: A single system that is in multiple pure states simultaneously, each of which contain all information about the system. Unless these can be rotated into a single pure state, their superposition provides an over-specification of the system.

Mixed State: A system that is in one of several distinguishable pure states and is described as a statistical combination of them.

Ensemble: A combination of distinguishable, identically prepared, noninteracting systems.

Pure and entangled states can be rotated in Hilbert space. By doing so, they can be decomposed into different Hilbert basis vectors. As long as the relative phases of their components are retained, their components can be physically separated and recombined (if the means exist) with no loss of information. When split into their components, the components of these states can interfere with each other.

Mixed and ensemble states represent mixtures of quantum states. These mixtures and ensembles obey classical statistics in the sense that expectation values for observables are calculated with respect to the same basis for all components. This means their component properties are separately and simultaneously measurable, though the component systems retain their quantum character.

“Technically, classical ignorance cannot be described as a simple quantum state, but rather requires a distinct treatment known as a mixture... The mixture describes a state of whose nature we are unsure—not for any reasons involving quantum-mechanical uncertainty, but simply because of classical ignorance... This is a very general point: *superpositions allow for interference, but mixtures do not.*”— **George Greenstein** (2006, p. 188)

The elements of mixtures and ensembles lack relative phase information and do not interfere. These elements either do not simultaneously exist in the same location or, if they do, they have been incoherently combined. Interference effects may be found within each system of a mixed state but not between the states of the mixture. Their probabilistic nature represents a lack of knowledge of the system rather than inherent quantum uncertainty.

Superposed states can lose the phase relationships of their components through a process of decoherence. When this happens, a superposition becomes a mixed state and the ability to create predictable, phase-induced interference between the state's components is lost. In addition, when we combine pure states, we can create either a superposition or a mixed state depending on whether or not we preserve the phase relationships between the components.

For example, a spin-1/2 particle in a spin-up state can pass through a left-right Stern-Gerlach machine and be decomposed into a superposition of spin-left and spin-right states. The spin-up particle is then described by a superposition of two components that retain their relative phase information. The original spin-up state is regained when the two phase-intact components are recombined.

This superposition only becomes a mixture when the relative phase information is lost. This happens through decoherence, such as when we create a situation where we can observe one of the two components. It is important to note that the phase relationships are not lost by the process of observation, but by the process of creating the situation in which they are lost, that is by our forcing them onto distinguishable paths, whether we perform the observations or not.

The point is that unless we additionally specify these phase relationships, the mathematical specification of the state does not indicate whether or not phase relationships are preserved. Hence, without the phases being specified we don't know whether a quantum state combination is a superposition or a mixed state. To make matters more complicated, we can also have quantum states that are combinations of superposed and mixed state components.

10 - Beyond Superpositions

The entangled states presented as describing EPRBA situations exhibit space-like separation between observations, nonlocal acausality, and are said to experience wave function collapse. After this collapse they exhibit path-specific nature.

Although the wave functions for each particle of the correlated pairs can be decomposed into pure state components—such as spin-up and spin-down—and each pure state can be represented by a superposition of orthogonal components—such as a combination of spin-left and spin-right components—the components of one particle does not interfere with the components of the other particle. They do not interfere because, after the wave function has collapsed, they follow specific paths and those paths are distinct.

In the case of a pair of fermions, each fermion of the correlated pair passes through a separate Stern-Gerlach machine. The two Stern-Gerlach machines are rotated relative to each other by a known angle in the plane perpendicular to the fermions' direction of travel.

In the case of photons, each photon of the correlated pair passes through a linear polarizing filter. The two linear polarizing filters are rotated relative to each other by a known angle in the plane perpendicular to the photon's direction of travel.

It is the path-specific requirement that is said to collapse the entangled wave functions. The observation of the resulting paths provides the measure of the pair's correlation.

In the fermion version of the EPRBA experiment we measure the paths the correlated fermions take upon exiting the Stern-Gerlach machines; in the photon version of the EPRBA experiment we measure the coincidence rate at which correlated photons are transmitted through the relatively oriented polarization filters.

Explorations of interference between a wave function's components require the existence of phases that interfere. As these component phases are lost, either through environmental decoherence or experimental measurement, the signature correlations that define the EPRBA paradox are also lost.

Although the so-called Bell states involved in the EPRBA situations do not require us to observe interference patterns, we continue to describe the states as superpositions of correlated, space-like separated components. Physical measurements on such states act to select one observable and exclude the other pure-state components. This selection is interpreted as an instantaneous collapse of the wave function into the observed state. This collapse is central to the paradox in that it begs an explanation of how space-like separated components enforce their correlations.

In the previous paper we trace the provenance of the supposedly entangled states used to confirm the disparity between observed quantum correlations and the limits given by the Bell-CHSH inequalities. There we show that the selection criteria used to identify and explore entanglement did not distinguish entangled from mixed states. Statistical differences do exist between the behavior of entangled and mixed states but these distinctions were not applied. As a result, to our knowledge, all of the “loophole tests” of the Bell-CHSH theorems could have involved mixed rather than entangled states. This does not mean that these tests were flawed as they unequivocally recorded quantum results, it’s rather that they did not distinguish between different quantum state configurations.

11 - Localizing Quantum Spin

In our previous study of the EPRBA situation for fermionic particles, we argued that the collapse of the superposed singlet wave function occurred within the time-like region at the source where the two particles initially overlapped. The particles then diverge from this region, moving in opposite directions.

This is an essential point: we are now working with a disentangled wavefunction: a wavefunction that is now a mixture of pure states. Having taken this step, we are no longer working in the same state space as those authors who derive the Bell-CHSH inequalities. Neither is this the same state space that’s presumed by those whose work confirms the results of experiments on entangled “Bell states.”

With that point established, we then showed that Bell-CHSH theorems exclude classical but do not prohibit non-classical local variables from reproducing the quantum results. We did this by relaxing Bell’s overly restrictive local factoring assumption and introducing a local $SO(3)$ vector variable λ representing quantum spin.

This local variable represents spin’s noncommuting properties. It is treated as a classical vector for the purpose of establishing a measure that allows us to integrate—in the sense of a path integral—all of the contributions that quantum spin makes to the observed correlation between detectors.

We need some analytic measure of the particles’ azimuthal spin contributions if we are to arrive at the correct predicted correlation for remotely correlated particles. Using this vector as a means of including spin’s azimuthal contributions enables us to obtain the observed angular dependency in the correlation formula.

Bell claimed this result could not be obtained with a classical theory, and this is not a classical theory. The $SO(3)$ vector λ that we’ve introduced allows us to include the effects of spin on the $SO(3)$ vector $\mathbf{a}$ that defines the orientation of our measurements, but the vector λ is not “real” in the sense of appearing in the expression for any observable, or being observed itself in any way. This is both a relaxation of Bell’s reality assumption, which is implicit in what we mean when we refer to a classical variable, and a realization of his objective of introducing an additional, local parameter. That is, our $SO(3)$ vector is local and 3-dimensional, but it is not classical.

12 - Local Factoring

Bell considered an expression $F(\mathbf{a}, \lambda)$ that gives the probability of observing a spin-up state when measuring the spin orientation using a Stern-Gerlach machine oriented an angle $\mathbf{a}$ from the vertical, which depends on a hidden variable λ . His factoring assumption is that F can generally be considered to be the product of a value function $A(\mathbf{a}, \lambda)$ —which only generates values $+1$ and -1 —and a weighting function $f(\lambda)$ —whose values range between 0 and 1. That is that $F(\mathbf{a}, \lambda) = f(\lambda)A(\mathbf{a}, \lambda)$.

Bell originally considered his factoring assumption to be a consequence of local causality (Brown 2016), but his is a specific and not a general assumption. That is to say Bell's factoring is not the only factoring that's consistent with his definition of locality, and he later suspected as much. We generalized Bell's factoring to allow functions f and A each of which depend on the angle $\mathbf{a}$ and the vector variable λ , that is $F(\mathbf{a}, \lambda) = f(\mathbf{a}, \lambda)A(\mathbf{a}, \lambda)$.

In our model the weighting function, $f(\mathbf{a}, \lambda)$, depends on the detector setting $\mathbf{a}$ and includes a similar weighting function relating the variable λ to the orientation of the second detector. Shimony, Clauser, and Horne (1976) noted the possibility of such a dependence in the weighting functions and objected to Bell's exclusion of this possibility. Extending their comments, Moldauer (1976) added that such an extension, “should cause no conceptual difficulties, and I believe that it permits us to restore local causality.”

These authors did not develop any model using this form and surmised that including such an interaction between the intrinsic variable λ and the randomly chosen detector orientations “would rely upon a metaphysics which has not been proved.”

It seems to us that coupling λ to the random value of spin is not more metaphysical that coupling it with the random orientation of the detectors. What's more, the use of such a coupling provides a deterministic mechanism that correctly couples the remote detectors. Our work partly confirms Moldauer's suspicion, but in our model, disentanglement was also required.

It could be argued that this form for $F(\mathbf{a}, \lambda)$ is not a “valid Bell-type hidden-variable model” which should avoid “any specific commitment about what physically exists and how it acts. (Norsen 2011)” This is true and it is exactly the point. We are not concerned with hidden-variable models that don't reproduce quantum results or don't specify local, causal mechanisms; we have developed a formulation that does both of these things. Our only concern with the Bell-CHSH inequalities is stay outside their limits in order to calculate results that are consistent with the observations.

In our model, λ is not a single-valued hidden variable, and $f(\mathbf{a}, \lambda)$ is not a finite polynomial function of it. As we'll see in section 14.3, the weighting function $f(\mathbf{a}, \lambda)$ has a value given by the vector inner product of its components, namely $\mathbf{a} \cdot \lambda$. Think of $f(\mathbf{a}, \lambda)$ as a weighting function which, when convolved with $A(\mathbf{a}, \lambda)$, generates a contribution whose sign is given by the projection of the polar component of the spin on the detector angle vector $\mathbf{a}$. This is developed explicitly in the following sections.

Our variable λ is an $SO(3)$ 3-vector representation of quantum spin which is a vector in $SU(2)$. It is not and cannot be treated as a simple 3-vector in 3-dimensional space, which is why it only appears in the

nonlinear functions $f(\mathbf{a}, \lambda)$ and $A(\mathbf{a}, \lambda)$. Neither of these functions are observable, and the expression for the correlation that's derived from them, which is observable, and no explicit λ dependence.

In our previous work we used this nonclassical, local, causal, hidden-variable construction to reproduce the correct quantum mechanical result for the correlation between spins. This result was considered impossible by those who believe Bell-CHSH theorems prevent the reproduction of the quantum result through the use of local variables. The key point is that it is not locality that prevents the observed quantum correlation, it is “classicality,” which is to say that it is the classical nature of the theory in which the variables appear.

Finally, we noted that this construction was, in fact, a version of non-relativistic quantum mechanics in which spinor degrees of freedom are handled using a path integral formalism with a well-defined measure. Which is to say that in this context normal, non-relativistic quantum mechanics is a local, causal theory.

In the current work we add a third possibility for recasting the superposition of photons as a mixed state. Rather than argue that the superposed components have experienced decoherence at the point of creation, which is when the bound state decomposes, we argue that the entangled wave function experiences decoherence and epistemic reduction due to the path-forcing choice of our measurement apparatus. That is to say, decoherence results from our preparation of the system for observation. It may be that these two points of view are the same in some sense—as is reflected in the delay-choice double-slit experiment (Zwirn 2017, Huang 2022)—but we do not explore that here.

13 - Simultaneous Mixtures

When we specify a component of the wave function as having a particular spin or polarization, we are specifying an observable result, not a preexisting intrinsic property of the object being measured. As we described in the previous work, assigning a particular 3-dimensional direction to quantum spin only tells us that the spin carried before observation lies on the forward-pointing unit-hemisphere of spin values, and that the axis of this hemisphere coincides with the specified eigenvalue's direction, which is the direction along which the measurement will be or has been made.

This hemisphere is half of a S^2 sphere (the 2-dimensional surface of a 3-sphere) in 3-dimensional space whose axis is aligned with the spin or polarization direction to be measured. The hemisphere defines the terminal points on the sphere, or unit Young polarization vectors on the Poincaré sphere. None of these vector values co-exist in three dimensions simultaneously with the value of spin (or polarization) vector, whose value is defined as a direction of the eigen basis in the Hilbert space.

That is to say, when we specify a spin along a positive z-axis in three dimensions, or a polarization on the Poincaré sphere, we are saying that only one 3-space direction is well defined. We can only simultaneously measure stationary spin values in directions parallel or antiparallel with the z-axis or the polarization axis.

We can specify a spin-1/2 particle's intrinsic spin as having one of two possible values but, unless or until we have prepared the particle in this way, we will not know what value the particle is carrying. Lacking this information, we cannot assign a spin direction to the particle. Instead, we assign a superposition of both. This a combination of our uncertainty of the binary direction, which is well defined but unknown, with our uncertainty of the latitudinal and longitudinal directions which are indeterminable, incommensurable with the direction given by z-axis eigenvector, and unknown.

The resulting symmetrized or anti-symmetrized wave function for a single particle α is:

$$\psi(\pm, \alpha) = \phi(+, \alpha) \pm \phi(-, \alpha)$$

Where $\phi(\pm, \alpha)$ indicates a particle α with a positive or negative spin along the z-axis, or a positive or negative polarization along the Poincaré sphere axis. For a correlated pair of particles this symmetrized or anti-symmetrized 2-particle wave function would be written as:

$$\psi(\pm, \alpha, \beta) = \phi(+, \alpha) \phi(-, \beta) \pm \phi(-, \alpha) \phi(+, \beta)$$

The essential point is that $\phi(+, \alpha)$ and $\phi(-, \alpha)$ represent the intrinsic state of the particle, while the addition or subtraction of the additional reversed component represent our ignorance. To put this another way, while the 2-particle wave function is a combination of two particles, it is a *mixture* of 2-particle states.

Despite these two 2-particle states being coincident in space and time, only one of these two 2-particle states exists. There is no interference between the two 2-particle terms because they are exclusive. We can call such a combination an “simultaneous mixture” to distinguish it from the superposition that describes the simultaneous and interfering presence of a wave function in each of the two arms of a Mach-Zehnder interferometer. What we've called a simultaneous mixture mixes potentiality with actuality.

“If the wave function is a real physical entity, its collapse implies a physical discontinuity in nature. If, however, it represents our knowledge, collapse simply reflects an update in information. Quantum mechanics itself does not clarify which of these perspectives is correct.” — Rajendra Kumar and Aram Singh (2025, p.43)

Because this 2-particle superposition is really a simultaneous mixture it does not experience a collapse. An observation merely tells us which of the two correlated arrangements existed before the measurement was made. The “reality” is that we have one or another local, 2-particle wave function component.

There remains the question of how the purely local wave function components, previously represented by the mixture, carry the remote correlation. The claim that local wave functions cannot reproduce the correct quantum correlations, motivated by the Bell-CHSH theorems, asserts this cannot be done.

This question is resolved for bosons in the same manner we previously resolved it for fermions, namely by showing that local correlations naturally result from local quantum variables when the photon's

internal spin is handled correctly. Bell-CHSH theorems present no obstacle because quantum spin is not “locally real” in Einstein’s sense.

This resolution of the EPRBA paradox involves three steps which, in logical order, are:

- 1- Recognition that spin or polarization specifications reflect prepared rather than intrinsic values.
- 2- Recognizing that the nonlocal wave function is a simultaneous mixture and, as such, does not physically collapse.
- 3- Demonstrating a local mechanism for the correlation of remote pairs.

In the previous work the step of localizing the wave function, given as step 2 above, was argued on the basis of decoherence at the source of their separation for fermions. We are going in the opposite direction here, starting with the nonlocal, entangled wave function and pointing out that its specification reflects what we have prepared for observation (Greenstein 2016, p.219ff) rather than a superposition that describes the system absent of such preparation.

14 - Correlation Between Detectors

We want to know the probability that a photon polarized in one direction, that is to say a photon that passes through a polarization filter oriented along a specific axis, will pass through a second filter whose axis is rotated by an angle θ relative to the first. The rotation is in the plane perpendicular to the photon’s direction of travel, and θ is the physical angle between the polarizers as set by experiment.

We can assume one detector (which selects the polarization of the particles that pass through it) is aligned with the vertical z axis. We can equivalently ask for the probability that this state of polarization, which is along the z -axis—which is known because the polarizer is oriented along the z -axis and the particle is first passing through this filter—will pass through the second filter which is rotated by an angle θ with respect to the first filter. Computing this correlation as a function of θ will give us the general relation for correlations between two filters as a function of their relative angles of rotation.

14.1 - The Nonlocal Quantum Result

Quantum mechanics provides an algebraic means of calculating correlations without reference to time, distance, or mechanism. We do this by simply projecting one state onto another.

We start with a state that represents the initial photons after they have passed through the filter at location A whose orientation is aligned with the z -axis. The second state, which we will project onto the first state, represents the same photons before passing through the first filter and after passing through the second filter that has been rotated with respect to the z -axis. The projection of the first state on the second, which is a rotated version of the first, gives the amplitude for photons to pass through both filters.

The photons are represented by the 2-dimensional spinor $|\Psi\rangle$. $|\Psi\rangle$ is rotated using the SU(2) rotation operator $R(\theta)$, as described in van der Warden (1974, p.92).

$$R(\theta) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -i \sin(\frac{\theta}{2}) \\ i \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}$$

If we define the polarization of the state that passes through either filter as positive and those that don't as negative, then we can represent the wave function's polarization as a combination of negative and positive components. We say that after it passes through the filter each photon is in a pure positive state, and if it is reflected from or absorbed by the filter it is in a purely negative state. We can write it as $|\Psi_+\rangle \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $|\Psi_-\rangle \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, where $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are the basis vectors in the eigenbasis defined with respect to the z-axis.

Defining the rotated positive state $R(\theta) |\Psi_+\rangle$ as $|\Psi'_+\rangle$, the probability amplitude for detecting the rotated positive state in the unrotated (z-aligned) positive state gives the amplitude that the photon will pass through both filters. This amplitude is:

$$\begin{aligned} \langle \Psi_+ | R(\theta) |\Psi_+\rangle &\equiv \langle \Psi_+ | \Psi'_+\rangle \\ &= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \cos(\frac{\theta}{2}) & -i \sin(\frac{\theta}{2}) \\ i \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \cos(\frac{\theta}{2}) \\ i \sin(\frac{\theta}{2}) \end{pmatrix} \\ &= \cos(\frac{\theta}{2}) \end{aligned}$$

Squaring the amplitude gives the probability as $\cos^2(\theta/2)$.

The probability amplitude for detecting the rotated positive state, $|\Psi'_+\rangle$, in the unrotated (z-aligned) negative state, $|\Psi_-\rangle$, is:

$$\begin{aligned} \langle \Psi_- | \Psi'_+\rangle &= \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(\frac{\theta}{2}) \\ i \sin(\frac{\theta}{2}) \end{pmatrix} \\ &= i \sin(\frac{\theta}{2}) \end{aligned}$$

The absolute square of this amplitude, $\sin^2(\theta/2)$, gives the probability of detecting the rotated positive state as a negative state.

The correlation for measuring the same states at both locations A and B is the probability of measuring both rotated states to be in the positive state minus the probability of measuring them in positive and

negative states. The correlation measuring positive states at A and B with a relative rotation of θ is defined as $E(+, A, B, \theta)$ where

$$\begin{aligned} E(+, A, B, \theta) &= \frac{(|\langle \Psi_+ | \Psi'_+ \rangle|^2 - |\langle \Psi_- | \Psi'_+ \rangle|^2)}{(|\langle \Psi_+ | \Psi'_+ \rangle|^2 + |\langle \Psi_- | \Psi'_+ \rangle|^2)} \\ &= \frac{(\cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2}))}{(\cos^2(\frac{\theta}{2}) + \sin^2(\frac{\theta}{2}))} \\ &= \cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2}) \end{aligned}$$

However, we're not focusing on either positive or negative states in particular, but only symmetric or anti-symmetric combinations. We want to include the correlations before and after rotation of the negative states. That is, we want to add, $|\langle \Psi_- | R(\theta) | \Psi_- \rangle|^2 - |\langle \Psi_+ | R(\theta) | \Psi_- \rangle|^2$ which generates the same probability as $E(+, A, B, \theta)$. Consequently, we multiply the numerator in the previous result by 2.

We normalize this result by dividing by the sum of the probabilities of all events, up and down, which is,

$$\begin{aligned} &|\langle \Psi_+ | \Psi'_- \rangle|^2 + |\langle \Psi_- | \Psi'_- \rangle|^2 + |\langle \Psi_+ | \Psi'_+ \rangle|^2 + |\langle \Psi_- | \Psi'_+ \rangle|^2 \\ &= 2 \left(\cos^2(\frac{\theta}{2}) + \sin^2(\frac{\theta}{2}) \right) \\ &= 2. \end{aligned}$$

Adding the separate correlations for positive and negative polarizations gives the combined correlation for symmetric and antisymmetric observed states as,

$$\begin{aligned} E(A, B, \theta) &= E(+, A, B, \theta) + E(-, A, B, \theta) \\ &= \frac{2(\cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2}))}{2(\cos^2(\frac{\theta}{2}) + \sin^2(\frac{\theta}{2}))} \\ &= \cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2}) \\ &= \cos(\theta) \end{aligned} \tag{1}$$

14.2 - Representing Photon Polarization on the Poincaré Sphere

The crux of the EPRBA paradox is the assertion that the correlated 2-particle spin states are represented by an entangled, superposed wave function. This was asserted by Bohm and Aharonov (1957) and seems to be accepted as fact. The assertion is incorrect, as we showed in the previous paper by calculating the invariance of the correlated particles' total spin under arbitrary rotations.

The entangled wave function suggested by Bohm and Aharonov is a mixture, not a superposition. It is not a representation of physical reality as they claim, but representation of what we don't know. Its components may be real, but the "collapse" is an epistemic phenomenon. To demonstrate that this is

the case, we show here that the information that establishes the correlation between photons is carried locally in their polarizations.

We want to calculate the probability that each of two photons, correlated so as to have net zero polarization, will each pass through a linear polarization filter when the axes of the two polarization filters, set in the same plane perpendicular to the photon's direction of travel, differ by an angle we'll now designate as ω .

Let $\mathbf{J}$ and $\mathbf{J}'$ be two opposite-pointing, unit 3-vectors $\mathbf{J}$ on the Poincaré sphere representing the polarizations of the two anti-correlated photons. In this case, $\mathbf{J}' = -\mathbf{J}$. These are not 3-space vectors they are Jones vectors, or qubits (Dür 2013). The $\mathbf{J}$ vector has only one stationary component defined by the direction in which $\mathbf{J}$ points on the Poincaré sphere. We denote this direction as the z-axis, and this component of the Jones vector as J_z . Jones vectors transform as spinors (del Castillo, 2013); their three components represent polarization not spatial direction.

Note that every pure, spin-1/2 electron (or spin-1 photon) wavefunction carries a spin (or polarization) J that is anti-correlated with its partner, for a pair whose total spin (or polarization) is zero. The directions of these vectors are always correlated, so any measurement of spin (or polarization) with respect to the same axis of measurement will yield exact anti-correlation. This result is not weakened by the integration we will do later in this section to calculate the weakening of the anti-correlation when measurements are made at unequal measurement angles.

In the semi-classical picture, the points on the surface of the Poincaré sphere represent the relative phases of the photon's two electric field components perpendicular to the direction of propagation. In the quantum picture, the Poincaré sphere represents the range of possible values of an internal quantum variable that correlates with the electric field but transforms according to $SU(2)$ and not $SO(3)$.

In the semi-classical picture, points around the sphere's equator represent polarization states where the two electric field values have real-valued, relative phase differences ranging from 0 to 2π , while points on the upper and lower half of the sphere represent complex-valued phase differences that range from $-i\pi/2$ to $+i\pi/2$.

All points on the surface of the Poincaré sphere can be specified using an azimuthal angle θ that ranges from 0 to 2π , and a polar angle ϕ that ranges from $-\pi/2$ to $+\pi/2$. Horizontal and vertical polarization are represented by antipodal points on the equator at $\theta = 0$ and $\theta = \pi$. Photons of $+45^\circ$ and -45° polarization are represented by equatorial points at $\theta = \pi/2$ and $\theta = -\pi/2$ and $\phi = 0$. Right and left circular polarization is located at the poles, perpendicular to the equatorial plane, at the polar angle values of $\phi = \pm\pi/2$.

Every point on the surface of the Poincaré sphere represents a polarization value. Any value can be transformed into another by a rotation of vectors on the sphere, and opposite polarization values always lie at antipodal points.

Note that orthogonal polarizations are represented by anti-parallel Jones vectors radiating from the center of the sphere, with a relative angle of π . In contrast, orthogonal vectors on the S^2 unit sphere in

physical 3-space are perpendicular and have relative angle of $\pi/2$. This is typical of both classical and quantum spinors and leads to the appearance of half-angle relationships when comparing spinor rotations with 3-D vector rotations.

While beams of photons can be unpolarized, which means the polarizations of the constituent photons are random, individual photons are always polarized. In the classical picture there is always some time invariant phase between a photon's perpendicular electric field components. In the quantum picture, every photon always has some value of internal "spin." We can never know the polarization of a given photon unless we have either prepared it, measured it, or measured another particle whose polarization is correlated with it.

We can assign a polarization value to a photon for calculational purposes, but we can never know a photon's polarization before measurement, and measurement entirely erases a photon's previous polarization. As such, unmeasured photon polarization can be considered a hidden quantum variable, something we can manipulate mathematically but to which we cannot assign a fixed value.

14.3 - Detector Correlation Based on Polarization Values

Define $A(\mathbf{a}, \mathbf{J})$, following Gu (2022), as the function that tells us whether we measure the photon at A either passing through or being reflected by a polarizing prism (Wyatt 1978, p.154) oriented at an angle given by a physical vector $\mathbf{a}$.

For linearly polarized light the vector $\mathbf{a}$ lies in the yz-plane perpendicular to the photon's direction of motion along the x-axis. For circularly polarized light $\mathbf{a}$ lies along the x-axis, with or against the photon's direction of travel. For elliptically polarization $\mathbf{a}$ lies off the yz-plane pointing in the forward or rearward direction.

The vector $\mathbf{a}$ has unit length and indicates a direction in physical space. $\mathbf{J}$ is the value of the Jones vector associated with the photon at A. The value $A(\mathbf{a}, \mathbf{J}) = +1$ indicates the photon is transmitted (passes through), while $A(\mathbf{a}, \mathbf{J}) = -1$ indicates that the photon is reflected or absorbed.

$\mathbf{J}'$ is assigned to the second particle as the negative of $\mathbf{J}$. We define a similar function $B(\mathbf{b}, \mathbf{J}')$ at B, such that $B(\mathbf{b}, \mathbf{J}') = B(\mathbf{b}, -\mathbf{J}) = -B(\mathbf{b}, \mathbf{J}) = A(\mathbf{b}, \mathbf{J})$. That is, B gives the negative of A for the same arguments. The function B indicates the photon's polarization measured at B using a machine oriented at the angle given by $\mathbf{b}$. Like $\mathbf{a}$, the vector $\mathbf{b}$ is also a vector on the S2 unit sphere.

The functions A and B only return the values ± 1 indicating polarization with respect to the measurement angles indicated by the vectors $\mathbf{a}$ and $\mathbf{b}$. A returns +1 when $\mathbf{J}$ projects along the direction of $\mathbf{a}$, and B returns -1 when $\mathbf{J}$ projects along the direction of $\mathbf{b}$. Let A and B return zero when the directions of their vectorial arguments are perpendicular.

As an ansatz, following Bell (1964, p.197), we define $A(\mathbf{a}, \mathbf{J})$ and $B(\mathbf{b}, \mathbf{J}')$ operationally as:

$$A(\mathbf{a}, \mathbf{J}) = \text{sign}(\mathbf{a} \cdot \mathbf{J}) \quad (2)$$

$$B(\mathbf{b}, \mathbf{J}') = -\text{sign}(\mathbf{b} \cdot \mathbf{J}') = -\text{sign}(\mathbf{b} \cdot -\mathbf{J}) = \text{sign}(\mathbf{b} \cdot \mathbf{J})$$

and $\text{sign}(0) = 0$.

Notice that when $\mathbf{a} = \mathbf{b}$, $A(\mathbf{a}, \mathbf{J}) = -B(\mathbf{a}, \mathbf{J})$ for all $\mathbf{a}$ and $\mathbf{J}$. This means the measurement of individual particles with respect to the same angle of measurement always results in exact anti-correlation.

This ansatz is substantiated using the following reasoning:

1. The polarization must have some representation in $\text{SO}(3)$ if its contributions to observables in $\text{SO}(3)$ are to be calculated in any form.
2. We know the polarization vector $\mathbf{J}$ cannot be represented by an observable 3-space vector since it has only one stationary component, whereas an observable $\text{SO}(3)$ vector requires three.
3. We know there are additional polarization contributions besides that given by the one observable polarization direction because, from the spinor algebra, $|\mathbf{J}|^2 > J_z^2$.
4. We know the two opposite polarizations are binary, antisymmetric, and exclusive (orthogonal).
5. We know the additional polarization contributions must be azimuthally symmetric because there is no observable azimuthal coordinate to break the “virtual” azimuthal symmetry.

The polarization vector $\mathbf{J}$ does not represent a direction of the electric field because $\mathbf{J}$ is a spinor, and spinors are not vectors. It is only the integrated value of $\mathbf{J}$ over the Poincaré sphere, multiplied by a weighting of that value of $\mathbf{J}$, that maps onto the electromagnetic polarization of the photon, and only in the context of a particular measurement. Through this process we obtain the correct quantum mechanical correlation between the detection of polarized photons, as we now show.

14.4 - Coupling Vectors to Spinors

Notice when $\mathbf{a} = \mathbf{b}$ the recordings at A and B are made along the same axes. In this case $A(\mathbf{a}, \mathbf{J})$ and $B(\mathbf{b}, \mathbf{J})$ are of opposite sign. This indicates the photons are anti-correlated.

Here $\mathbf{a} \cdot \mathbf{J}$ and $\mathbf{b} \cdot \mathbf{J}'$ are simple inner products of these pairs of 3-component objects. The function $\text{sign}(\cdot)$ returns the values $+1$ or -1 depending on the sign of these inner products. Equation (2) is written in terms of classical variables, treating all components of $\mathbf{J}$ as equally physical, and expresses the quantum result.

There is no classical explanation for equation (2) as it reflects the mapping of the properties of quantum $\text{SU}(2)$ spinors onto our classical $\text{SO}(3)$ space. $\mathbf{J}$ is represented by a vector for computational purposes, but it does not behave like a classical variable. Its behavior derives from the non-commutative properties of the Pauli matrices.

Because we never know and cannot measure the values of $\mathbf{J}$ or $\mathbf{J}'$ they must not appear in any physical result. We ensure the disappearance of $\mathbf{J}$ from our result by integrating over all values of it. Setting $\mathbf{J}' = -\mathbf{J}$ gives:

$$E(A, \mathbf{a}, B, \mathbf{b}) = -\int_{\Lambda} \rho_A(\mathbf{a}, \mathbf{J}) A(\mathbf{a}, \mathbf{J}) \rho_B(\mathbf{b}, -\mathbf{J}) B(\mathbf{b}, -\mathbf{J}) d\mathbf{J}$$

Since the density functions ρ_A and ρ_B are symmetric in $\mathbf{J}$ and are multiplied by the orientation functions $A(\mathbf{a}, \mathbf{J})$ and $B(\mathbf{b}, \mathbf{J})$, which are anti-symmetric in $\mathbf{J}$, we replace $\rho_B(\mathbf{b}, -\lambda) B(\mathbf{b}, -\lambda)$ with $-\rho_B(\mathbf{b}, \lambda) B(\mathbf{b}, \lambda)$ to get:

$$E(A, \mathbf{a}, B, \mathbf{b}) = \int_{\Lambda} \rho_A(\mathbf{a}, \mathbf{J}) A(\mathbf{a}, \mathbf{J}) \rho_B(\mathbf{b}, \mathbf{J}) B(\mathbf{b}, \mathbf{J}) d\mathbf{J} \quad (3)$$

Here $\rho_A(\mathbf{a}, \mathbf{J})$ and $\rho_B(\mathbf{b}, \mathbf{J})$ are the densities of $A(\mathbf{a}, \mathbf{J})$ and $B(\mathbf{b}, \mathbf{J})$ and Λ is the unit Poincaré sphere over which the single vector $\mathbf{J}$ is uniformly distributed.

The densities $\rho_A(\mathbf{a}, \mathbf{J})$ and $\rho_B(\mathbf{b}, \mathbf{J})$ are equal to the absolute value of the vector inner product of their arguments. The vectors $\mathbf{a}$ and $\mathbf{b}$ are of unit length. Their arbitrary angles reflect the measurement angles set by experiment.

$$\rho_A(\mathbf{a}, \mathbf{J}) = |\mathbf{a} \cdot \mathbf{J}| \text{ and } \rho_B(\mathbf{b}, \mathbf{J}) = |\mathbf{b} \cdot \mathbf{J}|$$

$\rho_A(\mathbf{a}, \mathbf{J}) A(\mathbf{a}, \mathbf{J}) = |\mathbf{a} \cdot \mathbf{J}| \text{ sign } (\mathbf{a} \cdot \mathbf{J})$, and this equals $\mathbf{a} \cdot \mathbf{J}$. $\rho_B(\mathbf{b}, \mathbf{J}) B(\mathbf{b}, \mathbf{J}) = |\mathbf{b} \cdot \mathbf{J}| \text{ sign } (\mathbf{b} \cdot \mathbf{J})$ which equals $\mathbf{b} \cdot \mathbf{J}$. Applying these to equation (3) gives,

$$E(A, \mathbf{a}, B, \mathbf{b}) = \int_{\Lambda} (\mathbf{a} \cdot \mathbf{J})(\mathbf{b} \cdot \mathbf{J}) d\mathbf{J}$$

This expression for the second moment of an isotropically distributed vector $\mathbf{J}$ is a standard vector-calculus expression. Rewritten in the spherical coordinates appropriate for integration,

$E(A, \mathbf{a}, B, \mathbf{b}) = \int_{\Lambda} (\mathbf{a} \cdot \mathbf{J})(\mathbf{b} \cdot \mathbf{J}) \sin\theta d\theta d\phi$, where integration over Λ means integration over the azimuthal angle ϕ = from 0 to 2π , and the polar angle θ = from 0 to π . Note that these angles exist in different spaces. The angles for $\mathbf{a}$ and $\mathbf{b}$ are in physical 3-space while the angles for $\mathbf{J}$ parameterize the Poincaré sphere. We are convolving identically dimensioned objects defined in different parameter spaces.

Writing $\mathbf{J}$ in rectangular coordinates as

$$\mathbf{J} = (r \cos\phi \sin\theta, r \sin\phi \sin\theta, r \cos\theta)$$

gives us the same expression we presented in the previous work, namely:

$$\begin{aligned} E(A, \mathbf{a}, B, \mathbf{b}) = & r^2 \left(\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (a_1 b_1 \cos^2\phi \sin^2\theta + a_2 b_2 \sin^2\phi \sin^2\theta + a_3 b_3 \cos^2\theta) \sin\theta d\theta d\phi + \right. \\ & + \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (a_1 b_2 + a_2 b_1) \sin\phi \cos\theta \sin^3\theta d\theta d\phi + \\ & + \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (a_1 b_3 + a_3 b_1) \cos\phi \sin^2\theta \cos\theta d\theta d\phi + \\ & \left. + \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (a_2 b_3 + a_3 b_2) \sin\phi \sin^2\theta \cos\theta d\theta d\phi \right) \quad (4) \end{aligned}$$

where a_i and b_i , $i = 1,2,3$, are components of $\mathbf{a}$ and $\mathbf{b}$. Equation (4) reduces to

$$E(\mathbf{A}, \mathbf{a}, \mathbf{B}, \mathbf{b}) = \frac{4r^2\pi}{3} (a_1b_1 + a_2b_2 + a_3b_3) = \frac{4r^2\pi}{3} (\mathbf{a} \cdot \mathbf{b})$$

$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos(\omega)$, where ω is the angle between $\mathbf{a}$ and $\mathbf{b}$. $\mathbf{a}$ and $\mathbf{b}$ are unit vectors so $|\mathbf{a}| = |\mathbf{b}| = 1$, and $\cos(\omega) \leq 1$. We can reset r , which is the length of $\mathbf{J}$, equal to $(3/4\pi)^{1/2}$ to “normalize” the correlation. The resulting expression for the correlation is,

$$E(\mathbf{A}, \mathbf{a}, \mathbf{B}, \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} = \cos(\omega) \quad (5)$$

Equation (5) is identical to the correlation given in (1) which was derived from quantum expectation values without any reference to hidden variables. Consequently, we see that it is the correlation of the internal angular momentum vector $\mathbf{J}$, which is both theoretically and practically unobservable, with the measurement angles given by $\mathbf{a}$ and $\mathbf{b}$ that explains the cosine relationship we observe.

There is no nonlocal effect between A and B. There is no physical collapse of the wave function. What appears as a collapse is a reduction in our ignorance regarding the spin state of the particle.

It is interesting to point out that while we think of polarization as an observable quantity it does not appear here, nor in any equation for any observable in any form. Quantum polarization is the z-component of the internal angular momentum to which, Saldanha (2020) argues, we cannot ascribe an objective reality even after measurements have been performed.

14.5 - Internal Spin, Magnetic Moment, and Polarization

In the previous work the electron’s internal spin aligned with its magnetic moment. In the current work the photons internal spin aligns with the Jones vector that represents its polarization. The reader might wonder at the disparity in claiming the magnetic moment carries the local information from which we establish remote correlation in one case, and the claim that the electric field polarization carries the local information in the other.

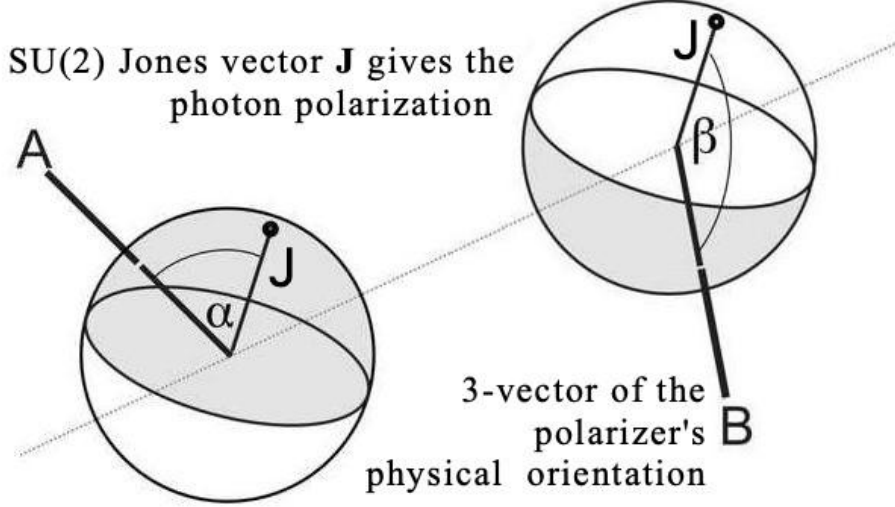
We remind the reader that internal spin is neither magnetic nor electric and has nothing to do with the classical picture of bar magnets or oscillating electric fields. It is the internal spin that carries the local information we use to establish remote correlations, and the internal spin is a quantum property that underlies quantum behavior in both cases. It is the quantum properties of spin that establish the correct correlations, not the classical field.

14.6 - Using J_z as a Hidden Variable

J_z defines an axis through the Poincaré sphere along which our two photons are polarized in opposite directions. This axis is defined by an angle in the Poincaré sphere. It is not a physical axis on S^2 , but we are putting it in coincidence with the physical axis along which we define the orientation of our

polarizers. We are using the angles on the Poincaré sphere as variable integration parameters that allow us to include the contribution of quantum polarization.

In the previous work we had the spin eigenvalue S_z whose axis coincided with the axis we used to



define the orientation of our Stern-Gerlach machines. We used the angles defining the spin directions as an integration variable to include the contributions of quantum spin components.

We can picture the intrinsic angular momenta as oppositely pointing Jones vectors subject to the caveat that only the z-component has a well-defined value. It might be better to picture the polarization variables as oppositely oriented hemispheres that have a specified z-direction and radius. In terms of Euler angles, the azimuthal angle θ of the positive polarization vector is uniformly distributed from 0 to 2π , while the polar angle ϕ is uniformly distributed between 0 and $\pi/2$. The azimuthal angle of the negative polarization vector is similarly evenly distributed from 0 to 2π , but its polar component ranges from $\pi/2$ to π .

*Figure 1: The unobservable directions of the photon polarization Jones vectors, imagined as the vector axes of two opposing hemispheres, and the unobservable angles α and β that lie between these axes and directions **A** and **B** in which the polarizations are measured.*

These hemispheres are an $SO(3)$, vector-based description of the $SU(2)$ spinor that describes each photon. The $SU(2)$ spinors have fixed z-components of unknown direction, while the $SO(3)$ vectors are constrained to either the upper or lower hemispheres.

Once we've established a correspondence between the physical direction of the detector and the Jones vector of the photon, we integrate over the unobservable Poincaré vector directions leaving a physical theory that only contains observable probabilities, which are the square of the transmission or reflection amplitudes due to the polarization.

We integrate over these physically nonexistent directions in the same way the path integral integrates over paths we cannot observe. If we attempt to "locate" the polarization components involved in this

integral through filtering, then we destroy the phase relationships that generate the phase correlation between the particles. The photon's internal polarization is the hidden variable that resolves the question of how polarization correlation is achieved.

This is analogous to the situation we encounter in the Young's Double-Slit experiment (Murray 2020) where we only measure one particle location but we must include all the possible paths the particle took to get there. Here, we will only measure one direction for the polarization but we must include contributions from all polarizations that satisfy the anti-correlation requirement. This is reminiscent of a sum over histories approach to calculating the partition function for quantum spin systems (Karchev 2012).

We never know nor can we ever assign a direction to the internal polarization before it has been measured. After it has been measured it is always transmitted or reflected with respect to the orientation of a polarization filter.

All we can observe are the directions A and B with respect to which the polarizations are measured, and the angles a and b between these vectors and the z-axis in the laboratory frame. The supposition that there exists a well-defined z-component of internal polarization before it has been measured is an unobservable construct.

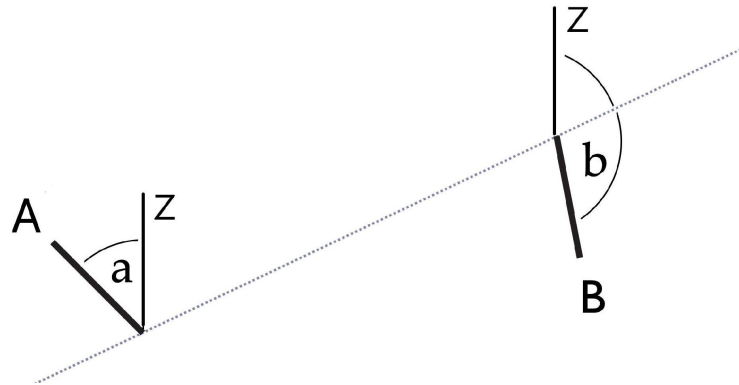


Figure 2: If when it's measured in direction A the first particle's polarization registers as parallel to A, then measured in direction B, whose angle is more than $\pi/2$ away from A, the second particle registers as parallel to B. We infer particle at A was polarized along the axis of the polarization filter, while the particle at B was perpendicularly polarized.

15 - Conclusions

A mixed state is the correct representation of two polarization-correlated photons, and it reflects our lack of knowledge of initial, unmeasured, intrinsic photon polarization. In this case, the wave function is epistemic in that its pure-state component reflects the state of the system, while its entangled structure represents our lack of knowledge.

Using nonentangled pure states presents the wave function as one of two nonstatistical, “real”—also known as ontological—models of the photons. To this we must “manually” add our ignorance

regarding which of the two possibilities the bound state decomposed into. When this ignorance is resolved through observation, we learn which of the two possible forms describes the state. In this case, what collapses is our ignorance, not the wave function.

This collapse of the paradigmatic example of quantum acausality—widely but misleadingly referred to as quantum nonlocality—suggests that quantum acausality does not exist. This poses the challenge of finding a local mechanism for correlations in other situations. For example, it implies that quantum teleportation (Pirandola 2015)—an effect that relies on nonlocal entanglement and lacks a causal mechanism—involves unrecognized mechanisms and is similarly misunderstood.

The difficulty of uncovering these mechanisms does not invalidate the necessity of finding them. It's taken since 1957, when Bohm and Aharonov presented their version of the EPR paradox, to produce the explanation given here. It would have taken less time if the absurdity of nonlocal acausality were recognized sooner.

Appendix I: Testing the Projection Postulate

If we identify the wave function as a complete description of an individual system, then we need to determine what happens to those parts of the system that instantaneously disappear, and what mechanism accounts for this.

Von Neumann (1932) asserted that observation reduces a superposed wave function to an eigenstate of the observation. This came to be called the phenomena of wave function collapse and was presented without further explanation. In the case of the EPRBA situation, where all wave function components are assumed to be physically real, the nonlocal nature of this collapse leads to acausal correlations between space-like separated observables, namely the correlation between observations at distant polarization filters.

We suggest that this wave function reduction of the entangled “Bell state” is explained by recasting the entangled state as a simultaneous mixture. Rather than a superposition of physical objects, the wave function is interpreted as a mixture representing an unknown combination. Mixtures do not collapse what is real, they only collapse our ignorance of it. The actual wave function does not change, and no physical projection occurs. This is testable by the following gedanken experiment.

Consider the measurements of the correlations of linear polarizations performed on two photons belonging to the same pair (Aspect 1982). Previous experiments involving such configurations have focused on verifying the violation of Bell's inequalities, but we are not concerned with Bell's inequalities here.

In the following proposed experiments, we focus on the effect of one of the two correlated systems on the state of the other when the two are separated by a space-like interval. In the conceptually simpler case of two correlated fermions, the experiment is shown in Figure 3. There, two particles with anti-correlated spin are generated in the source marked “S.” These particles travel in opposite directions toward two co-aligned Stern-Gerlach machines in the left and right regions, marked region A and B.

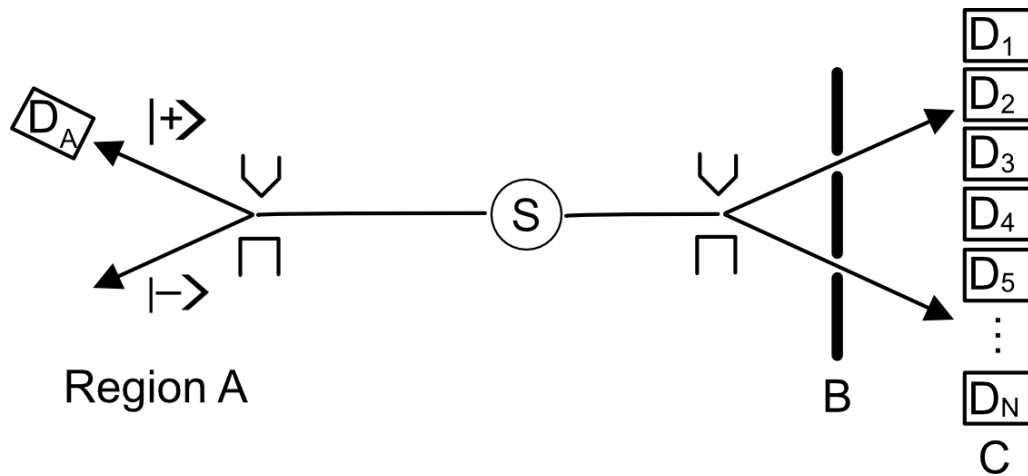


Figure 3: Combining two Stern-Gerlach machines and a double-slit apparatus. Does determining the path of the particle in Region A through the use of the detector at A affect the path of its correlated partner in Region B, and hence the interference pattern built up at the screen of particle detectors D_1 through D_N in Region C?

A particle detector is placed in the path of the higher beam in the left-ward region labeled A. A double-slit barrier is placed in right-ward region labeled B, and a screen of particle detectors is placed beyond the slits at location C.

We know that an interference pattern will build up on the screen at C only when we allow the wave function to propagate through both slits without measuring the presence of a particle at either of the slits. If the particle is recorded as passing through either slit, or if there is any way at all of recording through which slit the particle passed, then the double-slit interference pattern is replaced, over repeated trials, with the sum of two single-slit interference patterns, one generated from particles passing through the top slit and the other from particles passing through the bottom slit.

It is not necessary that one records which path the particle takes. It is sufficient to configure the apparatus so that this information is generated. Note that even a negative observation, the recording of no particle behind one of the slits, will determine the particle's path to have been through the other slit.

Because we know that interference between the components of a wave function can only occur when there is more than one component present in region C, we would like to know if the selection of one of two wave function components in region A simultaneously “collapses” the correlated wave function located in region C, separated from region A by a space-like interval.

Since the particles are spin-correlated, a particle that's recorded as traveling the upper path to the detector at A always correlates with a particle traveling in the lower slit at B. By coupling the particle detector at A with impacts on the particle detections at C we can determine whether or not recording the path detected at A affects the interference pattern that's built up at C.

If the correlation between particles is nonlocal, then a measurement in Region A will determine the path of the particle in Region C. If the conclusion of locality that's asserted here is correct, then nothing that's done in Region A will have any effect on what's observed in Region C.

The symmetry of this spin-1/2 particle experiment makes the possible nonlocal effect between regions A and B easy to visualize, but the nature of charged particles, the small dimensions involved, and the necessity of building up a spatial interference pattern in region C make this experiment difficult to realize.

An equivalent experiment involving photons is less symmetric and the test of nonlocal effects not as easy to visualize. However, the ease of directing, isolating, and recording light, the larger dimensions involved, and the simpler task of recording a temporal interference pattern (temporal beats) make this experiment easier to conduct.

The particle and the photon versions obey similar mathematics and test the same physics. Both can definitively test whether the wavefunctions of the correlated pairs are entangled or mixed, and whether or not a nonlocal collapse of the wavefunction occurs.

In the photon-based alternative, spatial alternatives of spin-up and spin-down particles reacting to an inhomogeneous magnetic field are replaced by oppositely polarized photons interfering when superimposed. The resulting single beam that enters the photomultiplier at D will or will not generate a temporal interference pattern due to a shift in the relative phases of the beam's two components. This experiment is shown in Figure 4.

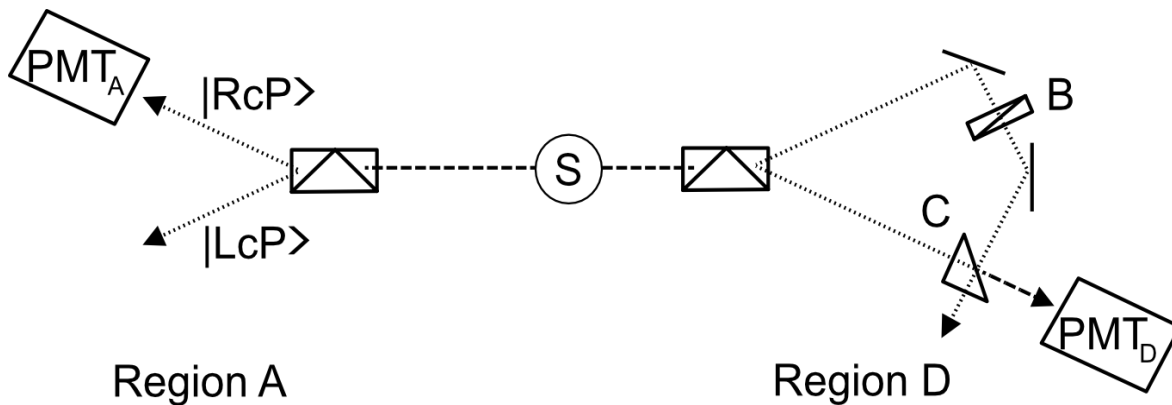


Figure 4: Combining two Fresnel polyprisms, which separate photons of left and right circular polarization, and an interference apparatus consisting of a $1/2$ -wave plate at B to shift the polarization, and a half-silvered mirror at C to recombine the beams. Does determining the path of the photon in Region A affect the path of its correlated partner and hence the temporal interference pattern recorded at the photomultiplier at D?

The experiment shown in Figure 4 uses Fresnel polyprisms to separate the correlated but unpolarized light generated at the source S into beams of right and left circularly polarized photons (Arteaga 2019). The experiment can be performed with single photons (Grangier 1986, 2005) or photon beams. If the conclusion of locality that's asserted here is correct, then, as in the previously suggested experiment,

nothing that's done in Region A will have any effect on the temporal interference pattern recorded by the photomultiplier in Region D.

The reason that Region A will not affect the observations in Region D is that the emitted photons are not prepared in any particular state. Their internal spins are correlated but their individual correlated values are unknown and, short of being prepared or measured, are unknowable.

The observation of a particular value in Region A forces a selection of one of two eigenvalues on the basis of the orientation of the detector at A. The result is determined by the spin or polarization of the photon at A. The second photon, on the opposite side of the source, is unaffected by this measurement.

The projection of the observed eigen value of the second particle is determined by the spin or polarization carried with this second particle to the detector at D. The photodetector(s) at D continues (or continue) to exhibit the usual interference pattern when the path of the second photon through the interferometer (or the double-slit apparatus) is unknown. A careful explanation of the detection of a photon passing through the interferometer in Figure 4 is provided by Loudon (2003).

Conflicts of Interest

The author declares no conflicts of interest regarding the publication of this paper.

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